



MINI PROJECT REPORT

Titled: Keyword extractor

Submitted By

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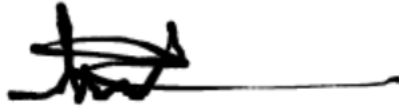
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DECLARATION

I declare that this project report entitled "**Keyword Extractor Using Natural Language Processing**" is my original work and has not been submitted to any other institution for academic credit. All sources of information used in this project have been duly acknowledged.

A handwritten signature in black ink, consisting of stylized, overlapping loops and a long horizontal stroke extending to the right.

Student Signature:

Date: 08/06/2026

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my lecturer for providing guidance and support throughout this project. I also appreciate my institution for providing the resources and learning environment necessary for completing this work. Finally, I thank my classmates and colleagues for their encouragement and valuable suggestions during the development of this project.

ABSTRACT

This project presents the design and implementation of a Keyword Extractor using Natural Language Processing (NLP) techniques. The system automatically identifies important words from textual content through preprocessing techniques such as tokenization, stopword removal, and frequency analysis. The project was implemented using Python and the NLTK library. The developed system successfully extracts meaningful keywords that represent the main concepts within a document. The project demonstrates how NLP techniques can be applied in information retrieval, document summarization, search engines, and content classification.

Keywords: Natural Language Processing, Keyword Extraction, NLTK, Text Mining, Information Retrieval.

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CHAPTER ONE: INTRODUCTION

1.1 Background of the Study

Natural Language Processing is a field of Artificial Intelligence that enables computers to understand, interpret, and generate human language. Organizations and individuals generate large amounts of text data daily through documents, websites, emails, reports, and social media platforms. Extracting useful information from this text manually is difficult and time-consuming.

Keyword extraction is an NLP technique used to identify important words and phrases that summarize the content of a document. These keywords provide a concise representation of the document and improve information retrieval processes.

1.2 Problem Statement

Many text documents contain large volumes of information, making it difficult for users to identify key concepts quickly. Manual identification of important terms is inefficient and prone to errors. Therefore, an automated keyword extraction system is required to improve efficiency and accuracy.

1.3 Objectives

General Objective

To develop a Keyword Extraction System using Natural Language Processing techniques.

Specific Objectives

- To preprocess textual data using NLP methods.
- To perform tokenization of text documents.
- To remove stopwords and irrelevant words.
- To identify important keywords from text.
- To display extracted keywords for user analysis.

1.4 Scope of the Study

The project focuses on extracting keywords from English-language text documents. The system uses frequency-based keyword extraction and NLP preprocessing techniques. Advanced machine learning and multilingual processing are not included.

1.5 Significance of the Study

The project contributes to efficient text analysis by enabling automatic identification of important information from documents. It can be applied in search engines, recommendation systems, digital libraries, and document management systems.

CHAPTER TWO: LITERATURE REVIEW

2.1 Natural Language Processing

Natural Language Processing combines Artificial Intelligence, Linguistics, and Computer Science to enable computers to process human language.

2.2 Keyword Extraction

Keyword extraction is the process of identifying significant words that best describe the content of a document. It is widely used in search engines, content classification, and text summarization.

2.3 NLP Techniques Used

Tokenization

Tokenization divides text into individual words or tokens.

Stopword Removal

Stopwords are common words such as "the", "is", and "and" that often carry little meaning and can be removed.

Frequency Analysis

Frequency analysis identifies words that appear most often in a document, helping determine important keywords.

2.4 Related Applications

- Search Engines
- Chatbots
- Recommendation Systems
- AI Assistants
- Information Retrieval Systems

CHAPTER THREE: SYSTEM ANALYSIS AND DESIGN

3.1 System Requirements

Hardware Requirements

- Computer or Laptop
- Minimum 4 GB RAM
- Stable Internet Connection

Software Requirements

- Python 3.14
- Jupyter Notebook
- NLTK Library

3.2 System Architecture

The system follows the workflow below:

Input Text → Tokenization → Stopword Removal → Keyword Identification → Output Keywords

3.3 System Flowchart

[Insert Flowchart Screenshot Here]

3.4 Algorithm Design

1. Accept text input.
2. Convert text into tokens.
3. Remove stopwords.
4. Count word frequencies.
5. Rank keywords.
6. Display results.

CHAPTER FOUR: SYSTEM IMPLEMENTATION AND TESTING

4.1 Development Environment

The project was developed using Python and implemented within Jupyter Notebook. NLTK was used for text preprocessing and keyword extraction.

4.2 System Implementation

Sample Code

```
import nltk
from nltk.tokenize import word_tokenize
from nltk.corpus import stopwords
from collections import Counter

text = "Natural Language Processing helps computers understand human language."

tokens = word_tokenize(text.lower())

stop_words = set(stopwords.words('english'))

filtered_words = [
    word for word in tokens
    if word.isalpha() and word not in stop_words
]

keyword_frequency = Counter(filtered_words)

print(keyword_frequency.most_common(10))
```

4.3 Testing and Results

Figure 1: Python Installation Verification

[Insert Screenshot]

Figure 2: Jupyter Notebook Environment

[Insert Screenshot]

Figure 3: NLTK Installation

[Insert Screenshot]

Figure 4: Keyword Extraction Processing

[Insert Screenshot]

Figure 5: Extracted Keywords Output

[Insert Screenshot]

4.4 Results Discussion

The system successfully identified meaningful keywords from the input text. The preprocessing stages improved keyword quality by removing irrelevant words and punctuation. The extracted keywords represented the central topics of the document accurately.

CHAPTER FIVE: CONCLUSION AND RECOMMENDATIONS

5.1 Conclusion

The project successfully implemented a Keyword Extraction System using Natural Language Processing techniques. The system demonstrated the ability to process text, remove unnecessary words, and identify meaningful keywords. The project highlights the practical importance of NLP in handling and analyzing textual information.

5.2 Recommendations

- Integrate machine learning techniques for improved accuracy.
- Support multiple languages.
- Develop a graphical user interface.
- Enable PDF and document processing.
- Implement phrase-based keyword extraction methods.

5.3 Future Work

Future improvements may include advanced keyword extraction algorithms such as TF-IDF, TextRank, and transformer-based NLP models to enhance extraction accuracy and performance.

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